

PROBLEM 1EXTENSION :

①

A rectangular aluminum bar (1.5×2.1 cm cross-section) and a circular steel rod (1 cm in radius) are each subjected to an axial force of 20 kN. Assuming that both are 30 cm long in their unloaded configuration.

Find: ① Stress in each structure (aluminum bar & steel rod).

② Extensional strain in each structure.

③ Amount of lengthening in each structure.

Let $E_{\text{Aluminum}} = 70 \text{ GPa}$

$E_{\text{steel}} = 200 \text{ GPa}$

$$\sigma = \frac{F}{A}$$

$$\epsilon = \frac{\sigma}{E}$$

SOLUTION:

$$\sigma_{xx}|_{AL} = \frac{20 \times 10^3 \text{ N}}{(0.015 \text{ m})(0.021 \text{ m})} = 63.5 \text{ MPa}$$

$$\sigma_{xx}|_{\text{steel}} = \frac{20 \times 10^3 \text{ N}}{\pi(0.01 \text{ m})^2} = 63.7 \text{ MPa}$$

$$\epsilon_{xx}|_{AL} = \frac{63.5 \text{ MPa}}{70 \text{ GPa}} = \frac{63.5 \times 10^6 \text{ Pa}}{70 \times 10^9 \text{ Pa}} = 907.0 \times 10^{-6} = 907 \mu\epsilon$$

$$\epsilon_{xx}|_{\text{steel}} = \frac{66.7 \text{ MPa}}{200 \text{ GPa}} = \frac{63.7 \times 10^6 \text{ Pa}}{200 \times 10^9 \text{ Pa}} = 318.3 \times 10^{-6} = 318 \mu\epsilon$$

- ② Assume the bar and rod are fixed at one end

unloaded length configuration

$$\delta = \int_0^L \frac{f}{AE} dx$$

Load

Young's Modulus

Cross-sectional area

$L = 30 \text{ cm} = 0.3 \text{ m}$

$$\begin{aligned} \delta|_{AL} &= \int_0^{0.3} \frac{20 \times 10^3 \text{ N}}{(0.015 \text{ m})(0.021 \text{ m})(70 \text{ GPa})} dx \\ &= \left[\frac{(20 \times 10^3 \text{ N})}{(0.015 \text{ m})(0.021 \text{ m})(70 \text{ GPa})} \right] \times \Big|_0^{0.3} \\ &= \frac{(20 \times 10^3 \text{ N})(0.3)}{(0.015 \text{ m})(0.021 \text{ m})(70 \text{ GPa})} = 272 \times 10^{-6} \\ &\Rightarrow 272 \mu\text{E} = \delta|_{AL} \end{aligned}$$

$$\begin{aligned} \delta|_{\text{steel}} &= \int_0^{0.3} \frac{20 \times 10^3 \text{ N}}{\pi (0.01 \text{ m}^2)(200 \text{ GPa})} dx \\ &= \frac{20 \times 10^3 \text{ N}}{\pi (0.01 \text{ m}^2)(200 \text{ GPa})} \times \Big|_0^{0.3} \\ &= \frac{(20 \times 10^3 \text{ N})(0.3)}{\pi (0.01 \text{ m}^2)(200 \text{ GPa})} = 95 \times 10^{-6} \\ &\Rightarrow 95 \mu\text{E} = \delta|_{\text{steel}} \end{aligned}$$

A human femur is mounted in a torsion testing device and loaded to failure. Assuming that one end is fixed and the other rotated, failure (fracture) occurs when $T = 180$ Nm and $\Theta(L) = 20^\circ$. Assume that $L = 37$ cm and that the failure occurs at 25 cm from the fixed end, where the inner and outer radii are 7 mm and 13 mm, respectively. Find the value of the shear stress at which failure occurs; estimate the shear modulus G .

$$\sigma_{\tau\theta}(r) = \frac{Tr}{J}$$

$$\sigma_{\tau\theta}(r=0.013m) = \frac{(180 \text{ N}\cdot\text{m})(0.013m)}{\pi(0.013^4 - (0.007^4))/2} = 56.9 \text{ MPa}$$

$$\Theta(z) - \Theta(0) = \int \frac{T}{JG} dz$$

$$\Theta(z=0.25m) = \frac{2(180 \text{ N}\cdot\text{m})}{\pi G (0.013^4 - 0.007^4)} \int_0^{0.25} dz$$

$$G = \frac{(360 \text{ N}\cdot\text{m})(0.25m)}{\pi \Theta(z=0.25m)(2.616 \times 10^{-8} \text{ m}^4)}$$

$$\frac{\Theta(z=L)}{0.37m} = \frac{\Theta(0.25)}{0.25m} \rightarrow \Theta(0.25) = \frac{20(0.25)}{0.37} = 13.51^\circ = 0.236 \text{ rad}$$

$$G = \frac{(360 \text{ N}\cdot\text{m})(0.25m)}{\pi (0.236 \text{ rad})(2.616 \times 10^{-8} \text{ m}^4)} \Rightarrow \boxed{G = 4.0 \text{ GPa}}$$